

SOUNDTRACE

Identify Any Music. From Any Video. Instantly.

- Music Recognition
- Spotify Links
- YouTube Links
- Trending Dashboard
- History
- Premium

Document Type	Complete Master Build Prompt — Full App Specification
Platform	Android (Primary) + iOS (Later) · Flutter Cross-Platform
Theme	White/Grey Textured Glassmorphism + Liquid Morphism Motion
Font Style	Bold Minimalist — iPhone System Bold Style
Core Technology	ACRCloud + AudD + Spotify API + YouTube API + yt-dlp
Monetisation	Google AdMob (Free) + ■99/month Premium Subscription

01

Design System & Visual Theme

SoundTrace uses a White/Grey Textured Glassmorphism theme — the only music app in India with this aesthetic. No dark mode. No parrot green. Pure premium white with frosted glass cards, subtle grain texture, bold iPhone-style typography, and living liquid morphism backgrounds.

COLOR PALETTE

Background White
Primary background — warm off-white, not pure white
#F7F7F5

Text Black
All headings, primary text — bold and sharp
#1A1A1A

Hairline
Borders, dividers, glass card edges
#DDDD9

Surface Grey
Card surfaces, input backgrounds, secondary areas
#EFEFED

Muted Grey
Secondary text, placeholders, metadata
#8A8A88

Pure White
Glass card fills, button text, highlights
#FFFFFF

TYPOGRAPHY

App Name / Hero	Helvetica Bold · 52–72pt · Tracking -2 · All Caps
Screen Title	Helvetica Bold · 28–36pt · Tracking -1 · Title Case
Section Header	Helvetica Bold · 16–20pt · Tracking 0
Body / Description	Helvetica · 13–15pt · Line Height 1.5 · Regular weight
Metadata / Labels	Helvetica · 11–12pt · All caps · Letter spacing +1
Button Text	Helvetica Bold · 14–16pt · Tracking 0 · Centre aligned

GLASSMORPHISM + LIQUID MORPHISM RULES

- Every card in the app uses frosted glass — white fill at 80–90% opacity with blur effect underneath
- Card borders are hairline grey (0.8px) with a white inner highlight on the top edge
- Drop shadows are soft grey (not black) — spread 12px, blur 24px, opacity 8%
- Background of every screen has animated liquid morphism blobs — slow moving, blurred, white/grey only
- The blobs in the background are 3–4 large soft circles that drift slowly — like oil in water
- On top of the blobs there is a fine grain/noise texture overlay at 4% opacity for the premium feel
- Floating music symbols (■ ■ ■ ■) drift very slowly in the background at 3% opacity
- All blob animations loop infinitely with easing — never jarring, always smooth and dream-like
- Glass reflections: each card has a subtle diagonal white shine from top-left to bottom-right
- No sharp corners anywhere — minimum 12px border radius on all elements

02

First Launch — Onboarding Flow

Hook welcome + feature walkthrough + premium upsell + Google login

SCREEN 0 — OPENING HOOK

When user opens the app for the very first time, they see a full screen with the SOUNDTRACE logo centred boldly in the middle. Below it, a single line appears letter by letter in a typewriter style — confident, not cold, not overly warm. Energetic and direct.

HOOK SCREEN COPY

Hook text appears letter by letter — typewriter animation

→ Line 1: " You just heard a song. You already forgot the name. "

→ Line 2: " Never again. "

After 2.5 seconds — a single CTA button rises from bottom

→ Button: [LET'S GO]

Style — bold, single line, centre of screen, no clutter

No images. No illustrations. Just bold text on the glass bg.

SCREENS 1-4 — FEATURE WALKTHROUGH SLIDES

SLIDE 1

Share Any Reel

Share directly from Instagram, YouTube Shorts, Facebook, or any app. One tap — your app opens automatically. No downloading. No gallery uploads.

Hint: Visual: Phone with Instagram open, share arrow pointing to SoundTrace icon

SLIDE 2

Music Identified In Seconds

Our engine recognises the exact song using audio fingerprinting — the same technology used by TikTok and major streaming platforms. Works even with voice-overs and background noise.

Hint: Visual: Sound wave animation resolving into a song card

SLIDE 3

Open Directly On Spotify or YouTube

One tap opens the exact song on Spotify or YouTube — not a search. The direct official link. Saves it to your history automatically.

Hint: Visual: Two buttons — green Spotify, red YouTube — side by side

SLIDE 4

Discover What's Trending In India

See the top 50 trending sounds in India updated daily. Filter by city, genre, and mood. Know what to use before it goes fully viral.

Hint: Visual: Trending list with city tags and growth percentages

SCREEN 5 — PREMIUM UPGRADE POPUP (appears on last slide)

On the final onboarding slide, a glass card slides up from the bottom (real physics spring animation). It does NOT block the content — it sits over the bottom 45% of the screen only.

PREMIUM POPUP SPEC

POPUP CONTENT:

→ **Headline:** " Go Premium. No Ads. No Limits. "

→ **Sub:** " Unlimited searches + Trending Dashboard + Ad-free "

→ **Price line:** " ■99 / month · or ■799 / year (save ■389) "

→ **CTA Button:** [START FREE 7-DAY TRIAL]

→ **Skip text:** " Continue with free version → "

ANIMATION: Card springs up with real bounce physics — overshoots slightly then settles

DISMISS: Swipe down to dismiss OR tap skip text

SCREEN 6 — LOGIN / REGISTER

- Full screen with SOUNDTRACE name centred bold in the middle of the screen
- Below the name: subtitle text — 'Your music. Your history. Always saved.'
- Three login options as glass pill buttons stacked vertically:
 - → [Continue with Google] — Google logo + text
 - → [Continue with Apple] — Apple logo + text (iOS only)
 - → [Continue with Phone] — OTP login via Firebase
- Below buttons: tiny text — 'By continuing you agree to our Terms & Privacy Policy'
- After login — user's name and photo synced from Google account into their profile
- All history, saved songs, and preferences tied to their account permanently
- Logging in on any device restores their complete history and settings

03

Core App Screens

Home · Song Result · Paste Link — The heart of the experience

SCREEN A — HOME

The home screen is clean and confident. The SOUNDTRACE name is bold and centred in the middle of the screen at all times — not at the top, centred vertically. Above it is the main action card. Below it is the paste link box and recent history.

Background	Animated liquid morphism blobs (white/grey only) + music symbol watermarks drifting slowly
App Name	SOUNDTRACE — Helvetica Bold 36pt — centred horizontally, positioned centre of screen
Main Card	Large frosted glass card with animated sound wave bars and text: 'Share a Beat to Identify Music'
Sub text	Below wave: Instagram · YouTube · Facebook · Any App with a Share Button'
Paste Box	Glass pill input below main card — placeholder: 'Or paste video link here...' + GO button inside
Recent	Recent labelled, 'See All →' right — then 3 glass song cards with wave art + name + time
Bottom Nav	4 tabs: Home · Trending · History · Profile — active tab has filled pill highlight

SCREEN B — SONG RESULT

After recognition the result screen appears with a smooth upward slide transition (real physics). SOUNDTRACE name remains centred on screen. All information is presented in clean glass cards stacked vertically.

RESULT SCREEN LAYOUT — TOP TO BOTTOM
TOP — Status badge: small pill → 'IDENTIFIED ✓' in grey on white
→ **SOUNDTRACE name — bold — centre screen**
→ **Album art placeholder — large glass card 200x200px — wave pattern inside (not copyrighted image)**
→ **Song name — Helvetica Bold 24pt — centred**
→ **Artist name — Helvetica 16pt grey — centred**
→ **Album · Year · Genre pill — small metadata row**
→ **30-second audio preview player — glass pill with play button and waveform progress bar**
→ **[OPEN ON SPOTIFY] — full width glass pill — Spotify green border + logo**
→ **[OPEN ON YOUTUBE] — full width glass pill — YouTube red border + logo**
→ **Paste / search box — glass input — 'Search another song or paste link...'**
→ **[Share →] [Save ♥] — two half-width glass buttons side by side**
→ **[← Search Another] — ghost text link at bottom**

TECHNICAL RECOGNITION FLOW — 4 LAYERS

MUSIC RECOGNITION — 4 LAYER SYSTEM
LAYER 1 — ACRCLOUD API
Send audio → searches 150M song database → returns song + Spotify ID + YouTube ID
Success rate: 85% of mainstream songs · Response time: under 2 seconds

LAYER 2 — AUDD API (BACKUP)
Send same audio with return=apple_music,spotify parameter
AudD searches its database + connects directly to Spotify → finds indie artists
Success rate: additional 8–10% of cases · Pricing: \$2–5 per 1,000 requests

LAYER 3 — YOUR OWN CROWD DATABASE (FIREBASE)
Every manually identified song saved with audio fingerprint
Next user sharing same audio → instant answer from your own database
Grows every day from user contributions — your unique competitive advantage

LAYER 4 — MANUAL LYRIC SEARCH FALLBACK
User types one word or lyric → Spotify Search API + YouTube Data API
Always returns something — user never hits a dead end

04

History & Trending Dashboard

Saved searches + India's paid trending music intelligence

SCREEN C — HISTORY

History screen shows all previously identified songs tied to the user's Google account. Because the account is linked, history is restored on any device. SOUNDTRACE name centred bold in the middle of the screen always visible.

Data Stored	Song name, artist name, genre, timestamp, city of search — TEXT ONLY
Not Stored	No audio files, no video, no album art images — 100% copyright safe
Art Display	Wave pattern placeholder in grey card — no real album images
Grouping	Entries grouped by date — Today / Yesterday / This Week / Older
Search	Search bar at top filters history by song name or artist in real time
Actions	Tap entry → reopens result screen with Spotify + YouTube links
Swipe	Swipe left on any entry to delete it from history
Clear All	Top right button clears entire history after confirmation dialog
Account Sync	History synced to Firebase — restored on re-login on any device
Privacy Note	Small banner: 'Only song names saved. No audio or video stored.'

SCREEN D — TRENDING DASHBOARD (PREMIUM PAID FEATURE)

The Trending screen is your premium monetisation feature. Free users see it locked with a blur overlay and upgrade prompt. Premium users see India's most searched sounds from your own app's data — updated every 24 hours, broken down by city and genre.

- Top 50 trending sounds in India — updated every 24 hours from your app's own user search data
- Each song card shows: rank number, wave art, song name, artist, search count, growth % this week
- VIRAL badge (orange) shown on songs with fastest growing search count
- Filter pills at top: All · Bollywood · Punjabi · South · Pop · EDM
- Regional filter: All India · North · South · West · East
- Rising Sounds section: songs gaining searches fastest — tagged 'Going Viral Soon ■'
- Tap any trending song → result screen opens with Spotify + YouTube links directly
- Weekly Trend Report: every Monday notification showing top 10 sounds of the week
- Free users see top 3 songs only — rest blurred with 'Upgrade to see all →' overlay
- Data is 100% from your own user base — not from Instagram API (which blocks this)
- How trending data is generated: every search logged anonymously with song + city + time
- Every 24 hours: backend counts logs per song → ranks by frequency → updates the list

TRENDING MONETISATION PRICING

Free users	See top 3 trending songs only · Updated weekly · No filter access
Premium ■99/month	Full top 50 · Daily updates · All filters · City breakdown · No ads
Premium ■799/year	Everything above · Weekly trend report PDF · Save ■389 vs monthly
Target earners	Creators (■149) · Businesses (■399) · Agencies (■799)

05

Profile Screen + Tech Stack + Monetisation

SCREEN E — PROFILE

- SOUNDTRACE name centred bold always at centre of screen
- User photo (from Google account) in circular glass frame — top of content
- User's display name and email pulled from Google login
- Subscription status pill: FREE or PRO badge
- Upgrade button (if free): glass card with trial CTA
- LINKED ACCOUNTS section:
 - → Link Spotify account — shows Spotify username when linked
 - → Link YouTube account — shows YouTube channel when linked
 - → When linked: future feature — import Spotify liked songs into history
- SETTINGS section:
 - → Notification preferences (weekly trend report on/off)
 - → Language: English / Hindi (future)
 - → Daily limit tracker (free users only) — shows X/5 searches used today
 - → Clear all history button
 - → Privacy Policy link
 - → Terms of Service link
 - → Contact / feedback button
 - → App version number at bottom
- LOGOUT button — glass pill at very bottom — logs out and returns to login screen

COMPLETE TECHNOLOGY STACK

Frontend	Flutter — single codebase for Android + iOS
Backend	Node.js or Python FastAPI — hosted on Railway.app (cheap + easy)
Database	Firebase Firestore — user accounts, history, crowd song database
Auth	Firebase Auth — Google login + Phone OTP
Music Recognition 1	ACRCloud API — 150M songs, 85% success rate, free tier 2,000/month
Music Recognition 2	AudD API — backup, \$2–5 per 1,000 requests, finds indie artists on Spotify
Audio Extraction	yt-dlp — extracts audio from Instagram/YouTube/Facebook shared links
Audio Processing	FFmpeg — trims extracted audio to best 20 seconds before sending to API
Spotify Links	Spotify Web API — free — search tracks, get official deep links
YouTube Links	YouTube Data API v3 — free — search videos, get official deep links
Ads	Google AdMob — banner + interstitial ads for free users
Payments	Razorpay — India payments: UPI, cards, net banking for subscriptions
Push Notifications	Firebase Cloud Messaging — weekly trend report alerts
Analytics	Firebase Analytics — track user behaviour and song search trends

MONETISATION MODEL

Google AdMob	Banner ad bottom of home screen + interstitial after every 3rd recognition
AdMob Revenue	India CPM ■30–120 · 50K daily users = ■45,000–1,80,000/month
Premium ■99/mo	Unlimited searches + trending dashboard + no ads + history sync
Premium ■799/yr	Same as monthly + save ■389 + weekly trend PDF report
Free tier limit	5 searches per day — after limit shows premium upgrade prompt
7-day trial	First time premium subscribers get 7-day free trial via Razorpay
Spotify Affiliate	Earn ■200–400 per Spotify Premium signup through your links
Music Labels	Paid partnerships to feature new releases in trending section

The Master Build Prompt

Copy and paste this into any AI coding tool or give to a developer

COPY THIS ENTIRE PROMPT → PASTE INTO AI TOOL OR SEND TO DEVELOPER

Build a mobile app called SOUNDTRACE for Android (Flutter).

■■■■ WHAT THE APP DOES ■■■■

Users share a Reel/Short/video from Instagram, YouTube, Facebook or any social app using the phone's native Share button. The app receives the video URL, extracts audio using yt-dlp and FFmpeg, sends it to ACRCLOUD for music fingerprinting, then shows the Spotify and YouTube deep links for that exact song. If ACRCLOUD fails, try AudD API with return=spotify. If both fail, show manual lyric search. Save every identification anonymously for the crowd trending database.

■■■■ DESIGN SYSTEM ■■■■

Theme: White/Grey Glassmorphism — NOT dark, NOT parrot green.

Background: Animated liquid morphism blobs (white/grey, slow drift)

+ fine grain noise texture at 4% opacity

+ floating music symbols (■■■) at 3% opacity drifting slowly.

Cards: Frosted white glass — 85% opacity white + hairline grey border

+ white inner highlight top edge + soft grey drop shadow.

Font: Helvetica Bold for all headings. Helvetica regular for body.

App name SOUNDTRACE: always centred in the middle of EVERY screen.

No sharp corners. Min 12px radius. Real physics spring animations.

Colour palette: #F7F7F5 bg · #1A1A1A text · #EFEFED cards · #8A8A88 muted

■■■■ SCREENS TO BUILD ■■■■

SCREEN 0 — HOOK: Full screen. SOUNDTRACE centred bold. Below it a typewriter animation reveals: 'You just heard a song. You already forgot the name.' pause 1.5s then: 'Never again.' Then a [LET's GO] button rises from bottom with spring physics.

SCREEN 1-4 — ONBOARDING SLIDES: 4 swipeable slides. Each has SOUNDTRACE centred bold at middle. Short bold headline. 2-line desc. Animated illustration hint at bottom. Progress dots. Skip button.

Slide 1: Share Any Reel. Slide 2: Identified In Seconds.

Slide 3: Open On Spotify or YouTube. Slide 4: Discover Trending India.

LAST SLIDE: glass card springs up from bottom — premium upgrade prompt:

'Go Premium. No Ads. No Limits. ■99/month or ■799/year. 7-day trial.'

CTA: [START FREE TRIAL] · dismiss: 'Continue free →'

SCREEN 5 — LOGIN: SOUNDTRACE centred. Below: Google login pill button

+ Apple login (iOS) + Phone OTP button. Firebase Auth. On login sync

user name and photo. All history tied to account permanently.

SCREEN A — HOME: Liquid morphism bg always animated. SOUNDTRACE name

centred vertically in screen. Large glass card above centre with sound

wave animation + text 'Share a Reel to Identify Music'. Below that: glass

pill input 'Or paste video link here...' with GO button inside. Below that:

Recent section with 3 mini glass song history cards. Bottom nav 4 tabs.

Free users see AdMob banner ad at bottom. Interstitial every 3 searches.

Daily limit: 5 searches free. After limit: premium upsell glass popup.

SCREEN B — RESULT: After share received — real upward spring transition.

SOUNDTRACE centred bold always. Stack from top: status badge 'FOUND ✓'

· album art glass card (wave pattern placeholder NOT copyrighted image)

· Song name Helvetica Bold 24pt centred · artist name grey · album+year+genre

· 30-sec preview audio player glass pill · [OPEN ON SPOTIFY] full width glass

pill (Spotify green border + logo) · [OPEN ON YOUTUBE] full width glass pill

(YouTube red border + logo) · paste/search box glass input · [Share] [Save]

two half buttons · [← Search Another] ghost link.

Deep links: spotify://track/{id} + youtube://watch?v={id}

SCREEN C — HISTORY: SOUNDTRACE centred. Search bar. Date grouped entries.

Each entry: wave art placeholder (NO real album art) + song name + artist

+ genre pill + time. Text only — copyright safe. Swipe left to delete.

Tap entry reopens result screen. Clear all button. Firebase sync so history

restores on any device after re-login.

SCREEN D — TRENDING (PREMIUM): SOUNDTRACE centred. Top 50 songs ranked

by search count in your app updated every 24 hours. Filter pills: genre + region.

Each card: rank · wave art · song name · artist · searches this week · growth %

· VIRAL orange badge if fast growing. Tap → result screen with links.

Free users: see top 3 only, rest blurred, upgrade overlay shown.

Trending data comes ONLY from your own app's anonymous search logs — NOT Instagram API.

SCREEN E — PROFILE: SOUNDTRACE centred. User photo glass circle + name + email.

PRO/FREE badge. Upgrade card (if free). Link Spotify account section.

Link YouTube account section. Notification settings. Clear history.

Privacy Policy + Terms links. Logout button at bottom.

■■■■ TECHNICAL REQUIREMENTS ■■■■

Frontend: Flutter. Backend: Node.js on Railway.app.

Database: Firebase Firestore. Auth: Firebase Auth (Google + OTP).

Audio extraction: yt-dlp → receives shared URL, downloads audio only.

Audio trim: FFmpeg → trim to best 20 seconds.

Recognition L1: ACRCLOUD API (free tier 2,000/month to start).

Recognition L2: AudD API with return=apple_music,spotify param.

Recognition L3: Firebase crowd database — community identified songs.

Recognition L4: Spotify Search API + YouTube Data API manual search.

Spotify API: search tracks + get deep links (free to use).

YouTube API: YouTube Data API v3 + get deep links (free to use).

Ads: Google AdMob (banner + interstitial for free users only).

Payments: Razorpay subscriptions (UPI + cards + net banking).

Anonymous trend logging: every search logs {song_id, city, timestamp}.